Investigating Voting Irregularities in the 2000 Presidential Election

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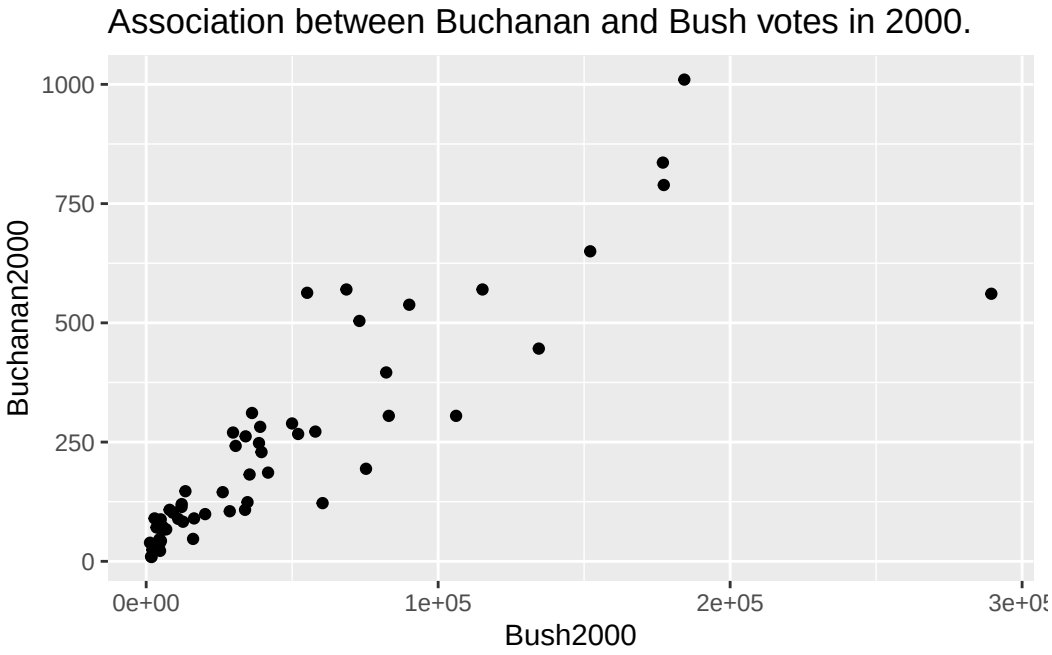
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1. Introduction

In the hard-fought U.S. Presidential election in the year 2000 between Republican Nominee George W. Bush and Democratic nominee Al Gore, the outcome hinged on the state of Florida. Because of a unique and confusing butterfly ballot system in Florida's Palm Beach County, many Democratic voters claimed their votes mistakenly went to Reform Party candidate Pat Buchanan instead of Al Gore. Voters' confusion with the ballot system is evident in the unusually large number of votes for Buchanan in Palm Beach County, as well as a striking number of 19,000 ballots discarded with two circles marked. This prompts the question: Did Buchanan receive an unusually large number of votes in Palm Beach County than in other counties, and if so, how many of those votes may have been intended for Gore? Using regression and prediction intervals, this case study aims to answer this question by comparing the voting pattern of other counties with that of Palm Beach County.

2.Data Description and Exploratory Data Analysis

The dataset includes Bush and Buchanan's vote counts in Florida's 2000 presidential election. Palm Beach County was excluded because its confusing butterfly ballot may have caused some Democratic voters to accidentally vote for Buchanan. The remaining 66 counties were used to show the usual relationship between the two candidates' votes. In these counties, Buchanan received a median of 111 votes, while Palm Beach County had 3,407 votes, much higher than any other county. This makes Palm Beach County an unusual outlier.



As shown in the scatterplot, the relationship on the original scale displays severe right-skewness and heteroskedasticity. Most data points cluster near the origin, and the variability in Buchanan votes widens substantially as Bush votes increase. This fan-shaped dispersion violates the constant variance assumption of ordinary least squares, necessitating a natural log transformation on both variables before fitting the regression model.

3.Modeling Process

Let $Buchanan_i$ denote the votes for Buchanan in 2000 in Florida county i, and $Bush_i$ denote the votes for Bush in 2000 in Florida county i. Our final linear regression model is

$$E[log(Buchanan_i)|log(Bush_i)] = \beta_0 + \beta_1 \log(Bush_i).$$

The estimated model parameters, along with their standard errors, test statistics, and p-values, are presented in Table 1 below. The estimated intercept is $\hat{\beta}_0 = -2.341$ (SE $= 0.354$), and the estimated slope is $\hat{\beta}_1 = 0.731$ (SE $= 0.036$). This positive slope indicates that, holding other factors constant, every 1% increase in Bush votes is associated with an estimated 0.731% increase in Buchanan votes across the baseline counties.

Table 1: Table 1: Linear Regression Model for Log(Buchanan Votes) as a Function of Log(Bush Votes) (Excluding Palm Beach County)

Term	Estimate	Std. Error	t value	p-value
(Intercept)	-2.341	0.354	-6.61	9.070663e-09
log(Bush2000)	0.731	0.036	20.32	1.300000e-29

4. 95% Prediction Interval for Predicted Buchanan votes in Palm Beach County

Assuming that the relationship between Bush votes and Buchanan votes was the same in Palm Beach County as it was in the other counties, a 95% prediction interval was created using the observed votes for Bush (152,846) in this county and the transformed fitted regression model for the 66 other counties. The resulting interval around the fitted value of 592.3769 (Buchanan's votes) was (250.8001, 1399.164).

5. Using the prediction interval to determine miscast votes

This suggests that with 152,846 votes for Bush, Palm Beach County was predicted to display approximately 593 votes for Buchanan with a 95% prediction interval ranging from about 251 votes to 1400 votes. However, the original data shows that Buchanan actually received 3,407 votes, which significantly exceeds the upper bound of the prediction interval. This implies that approximately 2,814 votes for Buchanan may have been miscast in this county.

Furthermore, the prediction interval can be used to determine the range of miscast votes by finding the difference between the observed votes and the upper and lower bounds of the prediction interval. This gives us a range of 2007 - 3156 for potentially miscast votes. Assuming these excess votes were intended for Gore, our findings suggest that the confusing butterfly ballot may have contributed to a substantial number of voting errors in Palm Beach County.

6. Limitations and Conclusion

In conclusion, our log-log regression model suggests that an unusually large number of votes may have been mistakenly cast for Pat Buchanan in Palm Beach County during the 2000 presidential election. The model estimates that about 2,815 votes may have been miscast, with a 95% prediction interval of 2,007 to 3,156 votes. This is much larger than George W. Bush's official 537-vote margin of victory in Florida. These results suggest that confusion caused by the butterfly ballot may have affected the election results.

However, our analysis has some limitations:

Generalizability limitation: Our model only uses county-level data from Florida in the 2000 presidential election. Therefore, the results may not apply to other states, elections, or ballot designs.

Causality and Association limitation: Our model shows a statistical relationship, but it cannot prove that the butterfly ballot caused the voting errors. We do not have direct evidence of what individual voters intended to select.

Model Assumptions: Although the log-log transformation helped address skewness and unequal variability in the data, some large-population counties may have had a strong influence on the regression results. Also, Palm Beach County may have differed from other counties because of its unique population and political preferences.

R Appendix

```
# Loading necessary packages
library(tidyverse)
library(Sleuth2)
library(broom)
library(kableExtra)

# Loading the data for case study one
election <- Sleuth2::ex0825

# Creating a second dataset with Palm Beach County excluded
election_wo_pb <- election |> filter(County != "Palm Beach")

# Compute summary statistics for contextual data description
summary(election_wo_pb$Bush2000)
```

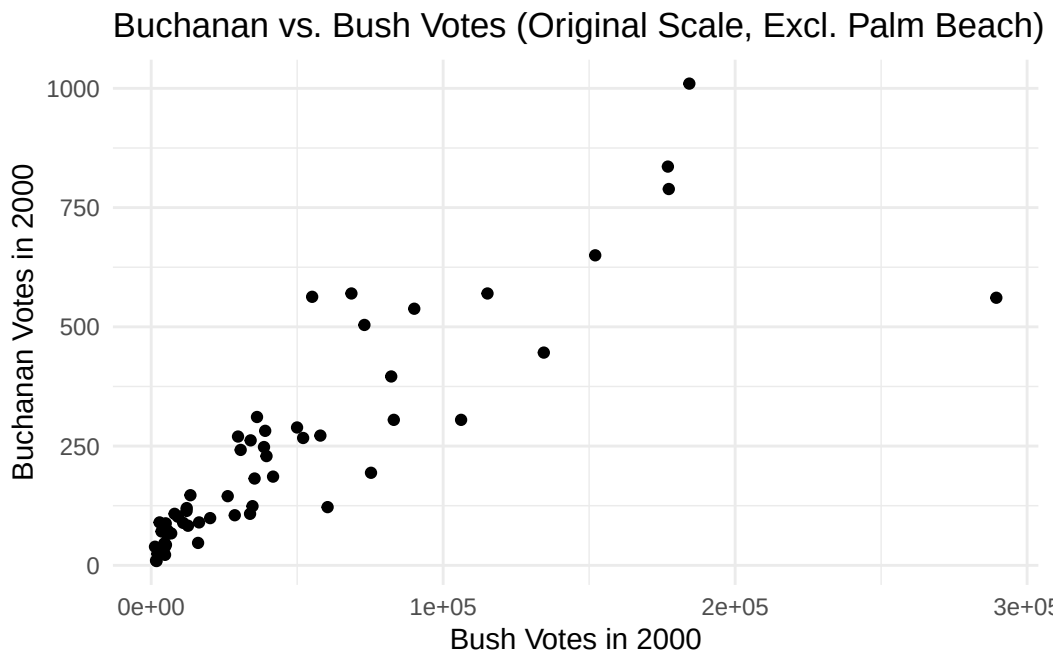
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1316    4745   18300   41697   54362  289456
```

```
summary(election_wo_pb$Buchanan2000)
```

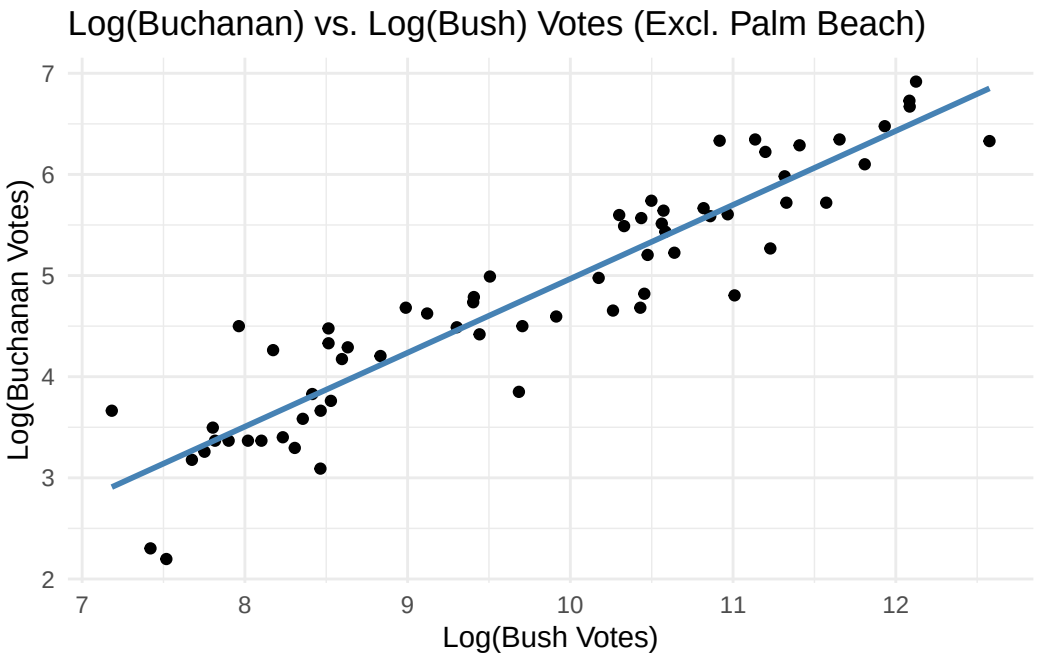
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   9.00   46.25  111.00  210.76  279.50 1010.00
```

```
# Creating a scatterplot for the relationship between Buchanan votes and Bush
 ↪  votes on original scale, without Palm Beach

ggplot(election_wo_pb, aes(x = Bush2000, y = Buchanan2000)) +
  geom_point() +
  labs(
    title = "Buchanan vs. Bush Votes (Original Scale, Excl. Palm Beach)",
    x = "Bush Votes in 2000",
    y = "Buchanan Votes in 2000"
  ) +
  theme_minimal()
```




```
# Scatterplot on log-log scale
ggplot(election_wo_pb, aes(x = log(Bush2000), y = log(Buchanan2000))) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "steelblue") +
  labs(
    title = "Log(Buchanan) vs. Log(Bush) Votes (Excl. Palm Beach)",
    x = "Log(Bush Votes)",
    y = "Log(Buchanan Votes)"
  ) +
  theme_minimal()
```




```
# Creating a readable regression table for the report
election_lm <- lm(log(Buchanan2000) ~ log(Bush2000), data = election_wo_pb)
election_lm_table <- summary(election_lm)$coefficients
election_lm_table |> kbl(col.names = c("Term", "Estimate", "Std. Error", "t
 ↪  value","p-value"), align = "c", booktabs = T, linesep="", digits = c(3, 3, 2,
 ↪  30)) |> kable_classic(full_width = F, latex_options = c("HOLD_position"))
```

Term	Estimate	Std. Error	t value	p-value
(Intercept)	-2.341	0.354	-6.61	9.070663e-09
log(Bush2000)	0.731	0.036	20.32	1.300000e-29

```
#. Palm Beach values
palm_beach <- election |>
  filter(County == "Palm Beach")

palm_beach
```

```
       County Buchanan2000 Bush2000
67 Palm Beach         3407   152846
```

```
# 95% prediction interval for Palm Beach

election_lm_pi <- predict(
  election_lm,
```

```
  newdata = palm_beach,
  interval = "prediction",
  level = .95)
pb_pi <- exp(election_lm_pi)
pb_pi
```

```
        fit      lwr      upr
67 592.3769 250.8001 1399.164
```